measurement shall be made on a corrugation tangent. Nominal Overall Thickness is Order Thickness.

[2] Minimum Base Metal Thickness shall be not less than 95% of the Design Base Metal Thickness. Minimum Base Metal is equal to the nominal Overall Thickness minus the Tolerance on Overall Nominal Thickness minus an allowance for zinc of 0.09 mm.

320.02.01.07 - Flatness tolerances shall apply to cut lengths only.

For specified thickness of 2.0 mm and thicker, the tolerance, or maximum deviation from a horizontal flat surface, shall not exceed 13 mm for specified width up to 1500 mm.

For thickness less than 2.0 mm the tolerance shall not exceed 20 mm for 900 to 1500 mm width, and 13 mm for less than 900 mm width.

320.02.01.08 - Permissible variations in width, length, and camber of flat sheet shall be in accordance with CSA G163.2.

320.02.02 Structural Plate Corrugated Steel Sheet and Plate - SPCSP sheet and plate shall conform to the relevant requirements of DBSS 320.02.01 except:

i) The maximum carbon shall be 0.10% and maximum manganese of 0.50%.

ii) Each lift shall be identified by the mill name or brand, test number, nominal mass, mill order number and dimensions either by marking the top plate of the lift or by a durable tag attached to the lift.

iii) Thickness and Tolerances shall conform to the requirements of Table 320-C.

320.02.03 Hardware - Hardware shall conform to the following requirements:

320.02.03.01 Rivets - Rivets shall conform to ASTM A 31, Grade A copper bearing, and shall be galvanized in accordance with ASTM A 153 Class D **or** ASTM A 164 Type RS.

320.02.03.02 Bolts and Nuts - Bolts shall be M20 x 2.5 nominal diameter and thread pitch as shown in Figure I. Bolts shall meet the chemical and mechanical properties of ASTM A 449. Nuts shall be as shown in Figure I and shall meet the requirements of ASTM A 563 Grade C. Bolts and nuts shall be galvanized to CSA G164 Class 5.

TABLE 320-C SPCSP SHEET AND PLATE BASE METAL THICKNESS AND TOLERANCE

Nominal Base Metal Thickness[1] (mm)	3.0	3.5	4.0	5.0	6.0	7.0
Design Base Metal Thickness[2] (mm)	2.84	3.37	3.89	4.95	6.0	7.0
Width of Blank (mm)	**SHEET TOLERANCES**[3] (mm)			**PLATE TOLERANCES**[3] (mm)		
1340	± 0.20	-	± 0.20	+0.40/-.030	+0.40/-0.30	+0.50/-0.30
1585	± 0.28	-	± 0.28	+0.50/-0.30	+0.50/-0.30	+0.50/-0.30
2315	± 0.30	-	± 0.30	+0.60/-0.30	+0.60/-0.30	+0.70/-0.30

[1] Nominal Base Metal Thickness is the specified Order Thickness. For field thickness check, add 0.09 mm to the Base Metal Thickness to account for galvanizing. Thickness shall be measured at a point on the sheet not less than 20 mm from a longitudinal edge. Field measurement shall be made on the corrugation tangent.

[2] Minimum Base Metal Thickness shall be not less than 95% of the Design Base Metal Thickness.

[3] Mill terminology.

320.02.03.03 Coupler Band Hardware - Nuts and bolts on coupler bands shall conform to ASTM A 307, Grade A and shall be galvanized according to ASTM A 153 Class C or ASTM A 164 Type RS.

Other hardware on coupler bands such as lugs and flanges shall be of material approved at the time the design of the coupler band is approved and they shall be galvanized in accordance with CSA G-164 Class 4.

If lugs or flanges are welded in place the weld shall be thoroughly wire brushed followed immediately by the application of a dry thickness of 50 μm of zinc rich coating in accordance with CGSB 1-GP-181M Coating, Zinc Rich, Organic, Ready Mixed.

320.03 Fabrication From Culvert Sheet

320.03.01 Corrugation Profile - CSP shall have corrugations which form smooth continuous curves in either helical or annular rings about the pipe axis with nominal inside diameters and profile dimensions as specified in Table 320-D. The profile radius of curvature (R) shall not be less than 50% of the depth of the corrugation as shown in Figure II.

320.03.02 Standard Sizes and Tolerances

320.03.02.01 Riveted CSP - Minimum outside circumferences for riveted CSP shall be as shown in Table 320-E

a) Measurement of minimum outside circumference may be made with a band which covers adjacent corrugation crests.

b) Dimensional Tolerances:

- **Diameter** - The average nominal inside diameter tolerance shall be ± 13 mm or ± 1%, whichever is greater, measured between inside crests of the corrugations perpendicular to pipe axis.

- **Length** - The specified individual pipe length shall be measured from the inside crest of the last corrugation at one end to the inside crest of the last corrugation at the other end. The overall length is greater than the specified length by approximately 50% of the corrugation pitch.

320.03.02.02 Helical CSP - Minimum outside circumferences for Helical CSP shall be as shown in Table 320-F.

320.03.02.03 Pipe Arch - Helical or Riveted - Standard pipe-arch dimensions measured from the inside crests of corrugations shall be as shown in Table 320-G.

TABLE 320-D CSP DIAMETER RANGE - CORRUGATION PROFILE (mm)

NOMINAL INSIDE RANGE (mm)	CORRUGATION PROFILE DIMENSIONS Pitch[1] x Depth[2] (mm)
100, 150, 200 and 250	38 x 6.5
200 to 2400 [3]	68 x 13
1200 to 3600	76 x 25
1200 to 3600	125 x 26

[1] Pitch shall be measured perpendicular to the direction of corrugations. Average pitch tolerance ± 3.0 mm.

[2] Depth shall be measured from outside crests to outside valley, with a straight edge laid across crests perpendicular to the direction of corrugation. Depth tolerance minus 5%.

[3] Flexibility Factor of concern for diameters greater than 2000 mm.

TABLE 320-E RIVETED CSP - MINIMUM OUTSIDE CIRCUMFERENCE

NOMINAL INSIDE DIAMETER (mm)	MINIMUM OUTSIDE CIRCUMFERENCE (mm)			NOMINAL INSIDE DIAMETER (mm)
	CORRUGATION DEPTH			
	13 mm	13 mm	25 mm or 26 mm	
200	680	3830	3900	1200
250	840	4460	4520	1400
300	1000	5080	5150	1600
400	1310	5700	5770	1800
500	1630	6330	6390	2000
600	1940	6950[1]	7020	2200
700	2260	7570[1]	7640	2400
800	2570		8580	2700
900	2880		9510	3000
1000	3200		10450	3300
			11380	3600

[1] Flexibility Factor of concern for pipes with this corrugation depth and diameters greater than 2000 mm.

TABLE 320-F HELICAL CSP - MINIMUM OUTSIDE CIRCUMFERENCE

MINIMUM OUTSIDE CIRCUMFERENCE (mm)							
Nominal Inside Diameter (mm)	CORRUGATION DEPTH (mm)						Nominal Inside Diameter (mm)
	6.5	11	13	11	13	25 & 26	
100	350			2900	2910	2970	900
150	500			3210	3220	3290	1000
200	660			3840	3850	3920	1200
250	820				4480	4550	1400
300		1000	1010		5110	5170	1600
400		1320	1330		5730	5800	1800
500		1630	1650		6360	6430	2000
600		1950	1960		6980	7050	2200
700		2260	2280		7610	7680	2400
800		2580	2590			8620	2700
						9560	3000
						10500	3300
						11440	3600

Note: Dimensional Tolerances:

Diameter - Average nominal inside diameter tolerance shall be ± 6.0 mm or ± 0.5% whichever is greater, measured between inside crests of the corrugations perpendicular to pipe axis.

Length - The length tolerance shall be ± 25 mm of the specified length. For CSP with recorrugated ends, the length tolerance for individual pipes shall be plus 50 mm minus 25 mm.

TABLE 320-G STANDARD PIPE-ARCH DIMENSIONS (mm)

SPAN	RISE	EQUIVALENT DIAMETER	MINIMUM CORNER RADIUS
450	340	400	75
560	420	500	75
680	500	600	75
800	580	700	85
910	660	800	100
1030	740	900	110
1150	820	1000	125
1390	970	1200	150
1630	1120	1400	175
1880	1260	1600	200
2130	1400	1800	225

Note: Span and Rise dimensional tolerance is ± 25 mm or 2% of equivalent round diameter, whichever is greater.

320.03.03 Riveted Construction

320.03.03.01 All rivets shall be according to the requirements of DBSS 320.02.03.01. All rivet diameters and lengths shall be not less than those shown in Table 320-H.

320.03.03.02 Fabrication Requirements

a) Hole centres shall not be closer than twice the hole diameter from any sheet edge, and shall be in reasonable alignment.

b) Rivets shall be driven cold without bending. Sheet edges shall be drawn tightly together throughout an entire lap length with rivet holes in reasonable alignment.

c) Rivets shall have full heads and shall completely fill the holes.

d) Longitudinal seams shall have at least one rivet for each corrugation, in the valley.

e) Double-riveting, or two rows of rivets along a longitudinal seam, shall be as in Table 320-I. The distance between centrelines of rows of rivets shall nominally be three times the hole diameter.

f) Circumferential seams shall have a maximum rivet spacing of 150 mm measured on centres, except that six rivets will be sufficient on 300 mm diameter or smaller.

g) Longitudinal seam lap shall equal or exceed minimums shown in Table 320-I. Circumferential seam lap shall be 30 mm minimum.

320.03.04 Helical Construction

320.03.04.01 Fabrication Requirements

a) The edges of the sheets forming the pipe shall be jointed with a continuous lock seam formed by folding over the edges of adjacent sheets so that they interlock with each other to form a tight joint. See Figure III.

b) The sheets shall be continuous throughout the length of the pipe, the joining of sheets by butt welding shall only be permitted provided the weld meets the following requirements:

- The butt joint shall be made by shielded electric arc process with a continuous deposit of material throughout the length of the joint. Penetration shall be

to the full depth of the material and no undercut, splatter or scale shall be permitted. If deemed necessary, sample of the joint shall pass a 90° bend test.

- The weld and adjacent area of damaged zinc shall be thoroughly wire brushed followed immediately by the application of a dry thickness of 50 μm of zinc rich coating conforming to CGSB 1-GP-181M Coating, Zinc Rich, Organic, Ready Mixed.

c) Ends of Helical CSP shall be finished with a straight cut perpendicular to the longitudinal centreline and any saw cut mismatch shall not exceed 15 mm. The ends shall be finished without significant burr. Cutting of previously manufactured stock to a shorter length with a hand held cutting torch is not permitted.

d) Standard fabricated lengths shall be in full metre lengths, as ordered. The length tolerance shall be ± 25 mm of ordered length. For CSP with recorrugated ends, length tolerance for individual pipes shall be plus 50 mm minus 25 mm.

320.03.04.02 Visual Inspection - by reference to Figure III the quality of the lock seam shall be judged by visual inspection. The pipe shall be cut so as to show the cross-section normal to the seam. The cut shall be of sufficient length to show the complete lock seam profile including offset.

a) The minimum seam lap dimensions shall be as shown in Table 320-J.

b) There shall be no roller indentations on either side of the 180º fold or beyond the lapped area.

c) The retaining offset as shown in Figure III shall be that portion of the sheet profile adjacent to the lock seam, and in tight contact with the 180º fold. The total retaining offset of both sides shall be a minimum of 1 t where t is equal to the nominal overall thickness of the sheet of which the pipe is constructed. The retaining offset on any one side shall be a minimum of 0.3 t. Figure III shows retaining offset of 1 t on both sides, for a total retaining offset of 2 t.

d) The 180º fold of metal at the lock seam shall have a smooth curvature both inside and outside with no angularity of the interior of the fold.

e) The gaps opposite the interior fold shall be equal on both sides of the seam, and each gap individually shall not exceed 1.0 t where t is equal to the nominal overall thickness of the sheet of which the pipe is constructed.

f) The helix angle (see Figure IV) shall be not more than 30° for helical CSP 600 mm diameter and greater.

320.03.04.03 Shear Strength of the Lock Seam - If the visual inspection required by DBSS 320.03.04.02 indicates that the joint is not tight for any reason, such as the lapping surfaces apparently not in intimate or complete contact, then the shear strength of the seam shall be tested by the test method defined in Appendix I. The minimum shear strength of the lock seam shall be 80 N/mm. If this test is performed, it shall be in addition to, not in lieu of, the visual inspection requirements in DBSS 320.03.04.02 and all requirements must be met.

320.03.04.04 Recorrugated Ends - Helical pipe ends may be recorrugated to provide two or four annular corrugations on each end of pipe for coupling purposes.

The visual checks and/or pull tests required by DBSS 320.09.04 Quality Control - Helical CSP shall (where possible) be carried out prior to recorrugation. Additional quality control pull tests will be required in the reformed end and the minimum lateral seam strength of the lock seam as per Table 320-K shall be not less than 60% of that required for the remainder of the pipe. There shall be no visible cracks in the base metal in the reformed end.

For Quality Management purposes, samples of the lock seam may be taken from the undamaged portion of a recorrugated pipe by means of a 38 mm hole saw. (See Appendix IV).

320.04 Coupler Systems

320.04.01 Purpose - Coupler systems for corrugated pipe shall resist water leakage, preserve pipe alignment, resist separation of adjoining pipe sections and resist root penetration or infiltration of normal backfill materials.

Note: When culverts are to be installed on steep grades, extra wide annular corrugated couplers are recommended for diameters greater than 600 mm.

Note: If recorrugated ends are specifically requested, then annular corrugated couplers (see DBSS 320.04.02) shall be provided.

TABLE 320-H RIVET DIAMETERS AND LENGTH (mm)

NOMINAL OVERALL CSP SHEET THICKNESS (mm)	NOMINAL RIVET DIAMETER (mm)		MINIMUM RIVET LENGTH (mm)
	CORRUGATION DEPTH		
	13 mm	25 mm or 26 mm	
1.0	8		16
1.3	8		16
1.6	8	10	16
2.0	8	10	16
2.8	10	12	20
3.6	10	12	25
4.2	10	12	25

TABLE 320-I RIVETED CSP - LONGITUDINAL SEAM LAP AND ROWS OF RIVETS

DIAMETER RANGE (mm)	CORRUGATION DEPTH (mm)	ROWS OF RIVETS	MINIMUM LONGITUDINAL SEAM LAP (mm)
200 to 500	13	Single	40
600 to 1000	13	Single	50
900, 1000	25	Double	75
1200 and larger	13 or 25	Double	75

TABLE 320-J HELICAL CSP - MINIMUM SEAM LAP DIMENSIONS

CORRUGATION PROFILE (mm x mm)	38 x 6.5	51 x 11	68 x 13	76 x 25	125 x 25
LAP (mm)	4.0	6.5	6.5	8.0	8.0

TABLE 320-K HELICAL CORRUGATED LOCK SEAM PIPE MINIMUM LATERAL SEAM STRENGTH

NOMINAL OVERALL THICKNESS (mm)	MINIMUM LATERAL SEAM STRENGTH (N/mm)
1.0	36
1.3	51
2.0	65
2.8	88
3.5	136
4.2	182
Greater than 4.2	234

320.04.02 Fabrication Requirements

320.04.02.01 Coupler bands shall be compatible with the conduit shape and corrugation profile, and designed so that it can be installed to lap equal portions of the pipes being connected.

320.04.02.02 Coupler bands shall be either corrugated or have circumferential rows of projections (i.e., dimpled). Both systems shall be designed to substantially mesh with the pipe corrugations and meet the requirements of DBSS 320.04.01.

320.04.02.03 Coupler standard dimensions shall be as shown in Table 320-L.

320.04.02.04 Coupler ends shall be drawn together about the pipe by bolts, wedges or other means of fastening to meet the requirements of DBSS 320.04.01.

320.04.02.05 Coupler flanges shall consist of rolled or formed angles or lugs (meeting the requirements of DBSS 320.02.03.03) of at least 3.2 mm effective thickness riveted or welded to the coupler.

320.04.02.06 - The minimum number of 12 mm diameter bolts (meeting the requirements of DBSS 320.02.03.02) per connection shall be two for the 180 and 300 mm coupler, three for the 600 mm coupler and four for the 900 mm coupler. Bolt length shall be not less than 150 mm.

320.04.02.07 Riveted & Weld Fasteners - Where flanges are riveted, the rivets shall have a minimum diameter of 10 mm (and shall meet the requirements of DBSS 320.02.03.01). The minimum number of rivets per angle shall be two for the 180 mm coupler, three for the 300 mm coupler, five for the 600 mm coupler and seven for the 900 mm coupler.

Where flanges or lugs are welded in place the weld area shall be thoroughly wire brushed followed immediately by two coats of zinc-rich coating conforming to CGSB 1-GP-181M.

320.04.02.08 Wedge Fastening - Coupler flanges shall be integrally formed in a manner to utilize fully the available areas of contact with the wedge. Lips of both coupler and wedge shall be formed to a contained angle of approximately 45º and shall be nominally 10 mm wide. Wedge shall be at least 16 mm high at the wide end suitable for hammering into position over the coupler and shall be at least 75 mm longer than the width of the couplers. Lips of wedges with engage lips of couplers shall be 13 mm minimum width. Wedges shall be no less than 2.0 mm thickness.

TABLE 320-L MINIMUM COUPLER DIMENSIONS

COUPLER TYPE	PIPE DIAMETER (mm)	NOMINAL OVERALL THICKNESS (mm)	MINIMUM WIDTH (mm)	NO. OF BOLTS	DIMPLES	
					MINIMUM NO. OF CIRCUMFERENTIAL ROWS	MINIMUM NO. PER ROW *
Flat	100-600	1.30	140	2		
Corrugated	300-600	1.30	180	2		
	700-1600	1.60	300	2		
	over 1600	1.60	300 or 600 **	3		
Universal	300-600	1.30	300	2	4	6
Dimple	700-1200	1.60	300	2	4	7
	Over 1200	1.60	600	3	4	8

* Minimum number of dimples per row is 6 for 76 mm x 25 mm corrugation profile.
** Width dependent on application.

320.04.02.09 Where gasketed joints are required for watertightness, they shall meet ASTM D 1056, ASTM C 361M, or requirements as stated in the Contract and Design.

320.05 Pipe Arch Corrugated Steel Pipe Construction

320.05.01 Pipe arch CSP is a multi radii shape having integral top, bottom and corners. The pipe arch shall conform to the dimensions shown in Table 320-G.

320.05.02 For riveted CSP pipe arch the staggered lapped longitudinal seam or seams shall be placed in the top arch section and not located in an area of change of radius or minimum radius.

320.05.03 - In the case of helical CSP pipe arch the minimum corner radii (see Table 320-G) apply also to the lock seams located at the corners; no transverse creasing of the seams will be permitted.

320.05.04 The longitudinal axis of the pipe centreline must be preserved straight, it shall not bend appreciably either vertically or horizontally. In the case of dispute the pipe arch shall be placed on a flat surface; the invert shall be in uniform contact with the flat surface throughout the length of the pipe and the inside of the pipe shall appear straight when viewed from either end.

320.06 Perforated Corrugated Steel Pipe

320.06.01 Helical and riveted perforated CSP shall meet all requirements of this specification for helical and riveted CSP before perforation.

320.06.02 The perforations shall be approximately circular and cleanly cut; they shall have a nominal diameter of 10 mm and shall be not less than 8.0 mm. The perforations shall be arranged in rows parallel to the axis of the pipe and shall be located in the outside valleys or along tangents of corrugations. Perforations are not required within 150 mm of each end of each length of pipe. The rows of perforations shall be arranged in two equal groups placed symmetrically on either side of a lower unperforated segment, corresponding to the flow line of the pipe. The spacing of the rows shall be uniform. The distance between the centrelines of rows shall be not less than 25 mm.

320.06.03 The minimum number of longitudinal rows of perforations, the maximum heights of the centrelines of the uppermost rows above the bottom of the inverts and the inside surface of chord lengths of the unperforated segments illustrated in Figure V shall be as specified in Table 320-M.

TABLE 320-M CSP INVERT PERFORATIONS

DIAMETER OF PIPE (mm)	MINIMUM ROWS OF PERFORATIONS	H[1] (mm)	L[2] (mm)
100	2	46	85
150	4	70	120
200	4	92	160
250	4	115	195
300	6	138	240
400	6	184	315
500	6	230	370
600	6	276	465

[1] **Maximum** Heights "H" of the Centreline of the Uppermost Rows above the bottom of the invert.

[2] **Minimum** Lengths of Inner Chord "L" of unperforated segments (See Figure V).

320.06.04 Flat band couplers shall be supplied with perforated pipe unless corrugated or dimpled couplers are specified. Couplers shall comply with all requirements of DBSS 320.04.

320.07 Structural Plate Corrugated Steel Pipe Construction

320.07.01 This Subsection covers the fabrication of SPCSP products, using sheet and/or plate as specified in DBSS 320.02. Shapes in addition to round include vertical and horizontal ellipse, pipe-arch, arch, vehicular, pedestrian, or animal underpasses, and others. Nominal inside diameters are given in Table 320-N.

320.07.02 SPCSP shall have annular corrugations. Crests and valleys shall form circumferential rings about a pipe axis. Profile (pitch x depth) shall be: 152 mm x 51 mm. Tolerances are given in Figure VI.

320.07.03 Standard Round - Nominal inside diameter for round pipe shall be as given in Table 320-M.

320.07.03.01 Structural plates shall be fabricated to standard dimensions shown in Table 320-O and Figure VII. Nominal corrugated plate lengths shall be 3050 and 3660 mm. Overall structure lengths shall be any combination of these two lengths, plus 100 mm.

320.07.03.02 Plates shall be punched along the edges with holes to form longitudinal and circumferential lap seams when assembled by bolting. Hole diameter shall not exceed bolt diameter by more than 6.0 mm.

320.07.03.03 The longitudinal hole pattern shall be a minimum of two holes per corrugation. Dependent on the seam strength requirements, three or four holes per corrugation may be specified (See Figure VII).

320.07.03.04 Plate arrangements in structures are normally designed with seams staggered such that not more than three plates come together at any one bolt hole.

TABLE 320-N SPCSP - NOMINAL INSIDE DIAMETERS (mm)

1500	2590	4610
1660	2740	4920
1810	3050	5230
1970	3360	5540
2120	3670	5850
2280	3990	
2430	4300	

TABLE 320-O SPCSP - FLAT PLATE WIDTH DIMENSIONS

WIDTH DESIGNATION NUMBER OF HOLE SPACES	NET ARC WIDTH (mm)	OVERALL OR BLANK WIDTH (mm)	NUMBER OF CIRCUMFERENTIAL BOLT HOLES
5N	1220	1340	6
6N	1465	1585	7
9N	2195	2315	10

Note: Number of circumferential bolt holes spaces (N) at 244 mm along plate net (arc) width. See Figure VII.

320.07.03.05 Plate arc width equals the number of circumferential bolt hole spaces (N) at 244 mm centres along the plate, in accordance with Table 320-O and Figure VII.

320.07.04 Plate Curvature

320.07.04.01 Plates shall be curved to a radius along the inner crest dimension (plate net arc width) to form the circular arcs as required for the shape specified.

320.07.04.02 Tolerance on calculated rise shall be ± 13 mm (See Figure VIII). Length of straight-edge shall be not less than 75% of chord length.

Note: Structural plate fabricators shall make available on request the standard rise dimensions for plates of varying curvature, using varying straight-edge inspection lengths.

320.07.05 Bolts and nuts shall be in accordance with DBSS 320.02.03.02

320.07.05.01 Bolt lengths shall be sufficient to provide full thread engagement in the nut when the assembled plates are tightened in place.

320.07.05.02 The number of bolts of each length furnished shall be plus 2% in excess of the theoretical number to field erect the structure.

320.07.05.03 Bolts and bolt containers shall be identified by length of bolt using colour codes given in Table 320-P.

320.07.06 Drawings and Plate Identification

320.07.06.01 SPCSP plates shall be adequately identified by fabricator for field erection. Marking shall be clear and distinct for each job or order number. Identification may include thickness, radius, and drawing or code numbering.

320.07.06.02 For each structure order the fabricator shall provide a bill of material. Erection drawing(s) shall be provided where required showing plate arrangements together with assembly instructions.

320.07.06.03 Cut plates for bevelled or skewed ends shall be legibly identified to designate positions in a finished structure, and referenced on erection drawings.

320.08 Asphalt Coated Pipe - This Subsection covers asphalt coatings applied to CSP products;

it does not cover pipe manufactured of asbestos bonded sheet.

Coating Designations:

AC - Asphalt coated
ACPI - Asphalt coated Paved Invert

320.08.01 AC - Asphalt Coated - The completed CSP manufactured in accordance with this specification shall be uniformly coated by a hot dip process with asphalt inside and outside to a minimum thickness of 1.3 mm each side measured on a corrugation crest. A minimum of two dips shall be required for each length of pipe. The first dip shall be prolonged sufficiently to ensure that the metal in the pipe is substantially at the same temperature as the asphalt, to ensure proper adhesion. The second dip shall be prolonged only sufficiently to build to the required thickness. No pin holing, bubbles or breaks in the asphalt coating shall be apparent after the second dip. During the cooling period after each dip the asphalt shall be protected from rain or snow and shall not be rolled or handled in any way to damage the coating.

TABLE 320-P SPCSP BOLT AND CONTAINER COLOUR CODE

BOLT LENGTH (mm)	COLOUR BOLT	COLOUR CONTAINER MARK	BOLT LENGTH (mm)	COLOUR BOLT	COLOUR CONTAINER MARK
30	None	White	50	Black	Black
40	Green	Green	80	None	Yellow
45	Red	Red	Nuts *	None	Blue

Note: Nuts may be shipped in Bolt containers

320.08.02 ACPI - Asphalt Coated Paved Invert - In addition to the requirements of DBSS 320.08.01 asphalt shall be applied to provide a smooth surface in the invert filling the corrugations for at least 25% of the circumference of a round pipe and 40% of the periphery of a pipe arch. Pavement shall have a minimum thickness of 3.0 mm above the crests of the inside corrugations, except where pavement edges feather into corrugations.

320.08.03 Coupler Bands - The coupler bands ordered with AC or ACPI pipe shall be coated as required by DBSS 320.08.01. Coupler band hardware shall not be coated.

320.08.04 AC Perforated Pipe - Shall be uniformly coated inside and out to a minimum of 0.8 mm per side. Not more than 15% of the specified perforations shall be covered with a film of asphalt. The diameter of the clear perforations shall be not less than 7.0 mm.

320.08.05 Coatings Generally - Surface of the asphalt coating immediately after solidification (and prior to any treatment with anti-stick material) shall have a smooth and shiny appearance free of pinholes and bubbles.

Asphalt coating shall not spall under normal handling within the range of maximum summer and minimum winter temperatures.

Asphalt coating sampled from the finished pipe shall be capable of meeting the Shock, Flow, and Imperviousness Tests as defined in Appendix II.

320.08.06 Handling - To prevent bruising, scaling, or scuffling of asphalt coated CSP, proper and careful handling shall be observed. Asphalt coated pipe shall not be dragged or pushed along the ground, or otherwise roughly handled.

The fabricator may use an anti-stick material such as soapstone, chalk, or sand to spray or dust on the finished pipe coating after cooling.

320.08.07 Storage - Prolonged storage in open sunlight of asphalt coated CSP is not recommended. Long exposure to sunlight may evaporate asphaltic volatiles, resulting in drying with alligator cracking. Asphalt coated CSP shall be installed and backfilled within a reasonable period after coating to ensure maximum protective benefits against hydraulic, abrasive, and corrosive conditions.

320.08.08 Repair - Spots where the asphalt coating has been damaged shall be repaired. The fabricator shall be responsible for the repair of any damage to

the asphalt coating preceding shipment. Repair may be made with cold asphalt mastic, provided the area is small. Redipping will be required for large areas.

320.09 Inspection, Quality of Work, and Quality Control

320.09.01 Inspection

320.09.01.01 The Contractor and the Ministry shall have access to the fabricating plant for inspection, and every facility shall be extended to the Contractor and the Ministry for this process.

320.09.01.02 Inspection may include checks on gauge thickness, weight of zinc coating, corrugation profile, outside circumference, quality of riveting or lock seam, and such other features as deemed necessary by the inspector and Quality Manager to meet this specification.

320.09.01.03 Sampling of the base metal for chemical analysis shall be made in accordance with ASTM A 444.

320.09.02 Quality of Work

320.09.02.01 In addition to compliance with the details of construction, the complete pipe shall show careful, proper quality of work in all particulars.

320.09.02.02 Among others, the following defects constitute poor quality of work, and the presence of any of them in a corrugated steel pipe may be cause for rejection: uneven laps, undue deviation from true shape, lack of rigidity, excessive variation from a reasonably straight centreline, ragged or diagonally sheared edges, loose, unevenly aligned or spaced rivets or bolts, poorly formed rivet heads of lock seams, or damaged lock seam metal, bruised, scaled or broken zinc coating, corroded or improperly cleaned and painted welds, illegible brand, significant dents or bends in the steel itself, and ragged or unfinished ends.

320.09.03 Repair of Damaged Zinc Coating

320.09.03.01 Small localized areas, scratches or burns in which the zinc coating has been damaged by welding or other fabrication procedures shall be repaired by careful wire brushing followed immediately by the application of a dry film thickness of 50 μm of zinc rich coating conforming to CGSB Standard 1-GP-181M Coating, Zinc Rich, Organic, Ready Mixed.

320.09.03.02 Gross uncoated areas shall be repaired by re-galvanizing by hot-dip process in accordance with DBSS 320.02.01 or by metallizing process which shall meet the requirements of CSA Standard G-189.

320.09.04 Quality Control Helical CSP - The supplier shall develop and maintain an effective quality control system to ensure that adequate inspection coverage is maintained throughout the manufacturing process. Evidence of such inspection shall be available to the Quality Manager and the Ministry prior to shipment from the place of manufacture or from the supplier's storage facility.

The Contractor shall, and the Ministry reserves the right to, audit the supplier's quality control systems to ensure conformity with this specification.

The supplier shall furnish the Contractor with an outline of the supplier’s quality control procedures, for inclusion in the Quality Management Plan. The outline shall include the intended method of identification of production runs, frequency of visual checks and pull tests, method of reporting and recording, and/or retention of samples.

320.09.04.01 Production run shall be considered to include all pipes of the same metal thickness and diameter produced continuously, with only minor adjustments to the machinery being required. A new production run shall be considered to start with a change in metal thickness or a change in pipe diameter.

320.09.04.02 Visual checks shall at least include observations or measurement, as applicable, of metal thickness, corrugation depth, diameter, length, square end cut and zinc coating. They shall also at least include observation or measurement, as applicable, of the lock seam as follows: lap, gap, retaining offset, loss of seam contact, interior angularity, and roller indentation.

320.09.04.03 Pull tests shall be conducted in accordance with Appendix III. The lateral seam strength shall meet the minimum requirements of Table 320-K.

If any sample fails this requirement, all pipe already

produced in that production run or all pipe produced since the last pull test was conducted shall be subject to rejection. The manufacturer may however conduct additional pull tests on the previously produced pipe and thereby ascertain which, if any, can be considered acceptable.

To be considered acceptable a lock seam must meet all visual criteria as well as the minimum lateral seam strength requirement.

SECTION 320

CORRUGATED STEEL PIPE

APPENDIX I

Test Method for Shear Strength of the Lock Seam of Helical Pipe

(Referred to in DBSS 320.03.04.03)

A1.1 Apparatus

a) Testing Machine - Any standard testing machine capable of maintaining the specified rate of strain and of measuring the tensile load with an error not to exceed ± 50 N.

b) Grips - The grips for holding the test specimen in the testing machine shall be of the self-aligning type.

A1.2 Test Specimens

a) Sample - A sample shall be taken from the end of a completed pipe. The sample shall be cut out with a metal cutting saw, the cuts being made at least 12 mm on either side of the seam. The length of the sample measured along the shorter side of the seam shall be at least 255 mm. The end cut shall be made at right angles to the direction of the seam.

b) Specimen - Three separate specimens, each 75 mm long, shall be cut from the sample commencing at the end of the sample opposite the end of the pipe. See Figure 1. The remaining piece of seam at the end of the pipe shall be discarded. Each specimen shall then be prepared as shown in Figure 2. Extreme care shall be exercised in preparing the specimen to ensure that the seam is not loosened. If prepared in a vise it shall be held and cut such than no forces are applied which will tend to warp, twist or flatten the specimen.

A1.3 Procedure

a) The gauge length of each specimen shall be measured with calipers to an accuracy of ± 0.2 mm.

b) Each test specimen shall be carefully placed in the grips of the testing machine in such manner as to ensure that the axis of the test specimen coincides with the direction of the applied pull.

c) The load shall be applied at the rate stated in Appendix A1.4 until the test specimen has slipped 6.0 mm. The maximum load in Newtons shall be recorded.

A1.4 Speed of Testing

The load shall be applied continuously throughout the test at a uniform rate of motion of the moveable cross head of 2.5 mm per minute.

A1.5 Calculation

The shear strength of the lock seam of each specimen shall be calculated by the following formula:-

$$S = P/L$$

Where: S = Shear strength of lock seam, Newtons per lineal millimetre.
P = Maximum load obtained within first 6.0 mm of slippage, Newtons
L = Gauge length, millimetres.

A1.6 Report

The shear strength of the three specimens shall be reported and the average shall be the shear strength of the sample.

A1.7 Acceptance

A test sample shall be considered to meet the test requirements provided that the above average shear strength exceeds the requirements of 80 N/mm required by DBSS 320.03.04.03 and provided that at least two of the specimens also exceed that requirement. Should any test sample

fail to meet the test requirements, the manufacturer shall be allowed a retest on two additional samples for each sample that failed. Both retest samples are required to meet the test requirements before the pipe can be considered acceptable.

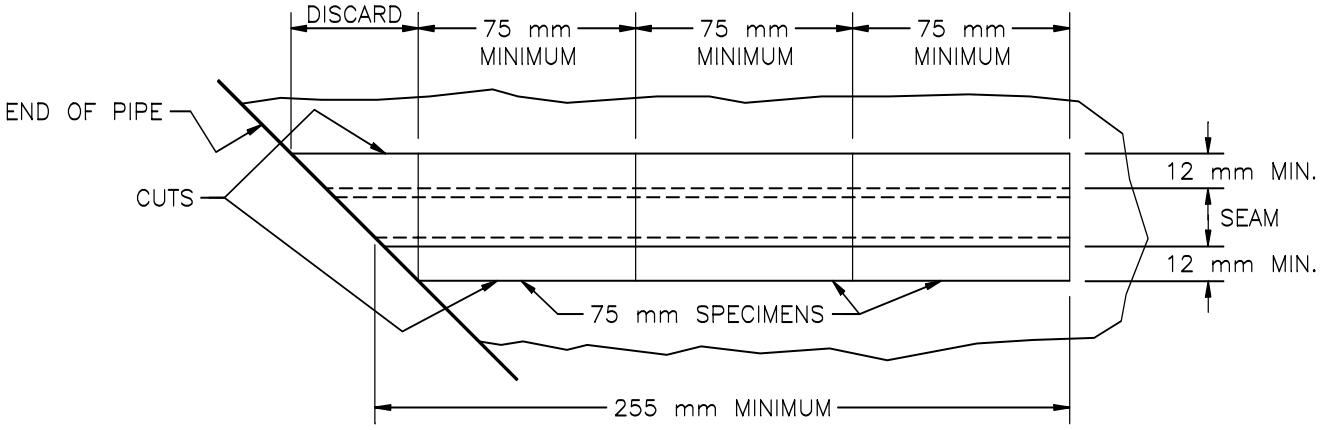


FIGURE 1 - DIAGRAM FOR CUTTING SEAM SAMPLE FROM END OF HELICAL PIPE

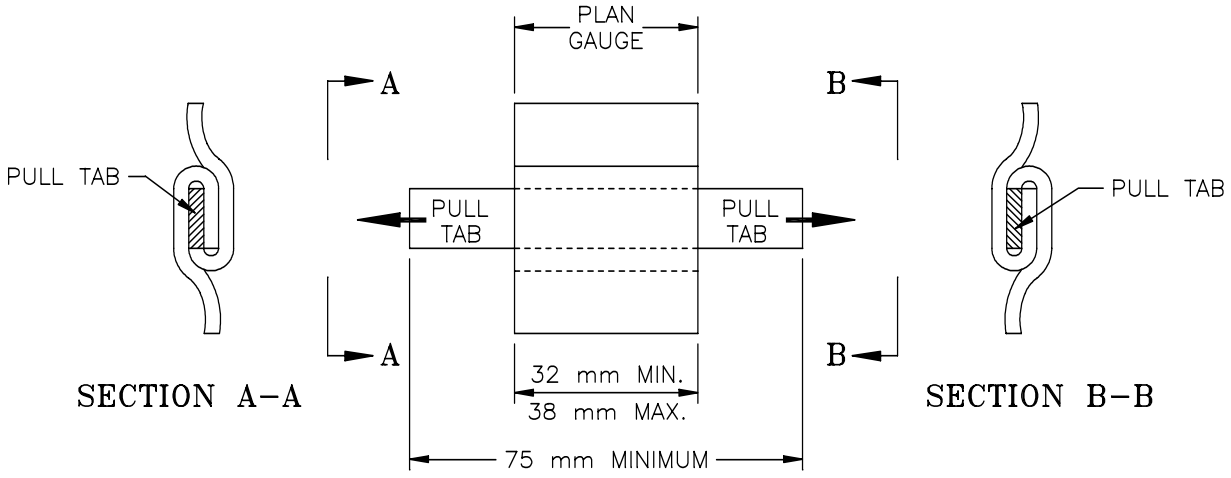


FIGURE 2 - DETAIL OF SPECIMEN FOR SHEAR STRENGTH TEST OF LOCK SEAL

SECTION 320
CORRUGATED STEEL PIPE

APPENDIX II

ASPHALT COATING
(Referred to in DBSS 320.08)

A2.1 Coating

The asphalt used for the AC and ACPI pipe shall have the following properties:-

	Limit %	Test Method
Asphalt Cement with no additives		
Loss on heating to 163°C, not more than	1	ASTM D 6
Penetration of residue after heating compared with penetration of same sample before testing for loss on heating, not less than	85	ASTM D 5 & D 6

A2.2 Sampling

All tests on the asphalt coating shall be made on samples obtained from the pipe on or about to be delivered to the project.

A 150 g sample of asphalt coating shall be obtained by gathering strippings from the inside top of one or more lengths of pipe. (Care should be taken in sampling to avoid contamination from sand or soapstone that may have been applied after dipping).

A2.2 Shock Test

The ability of coated pipe to withstand handling in cold weather is indicated by successfully passing the following shock tests:

a) Apparatus: The apparatus for the shock test shall consist essentially of a rigid metal anvil or base plate not less than 12 mm thick and resting on a solid foundation, and a hammer weighing 2 kg arranged to fall freely in suitable guides against a plunger weighing 1 kg and sliding freely in a vertical sleeve. The lower end of the plunger shall be hemispherically shaped, with a radius of 12 mm. The hammer shall be held in its guide by a trip mechanism at a distance of 130 mm above the plunger. Split molds for preparing the specimens shall be made of brass with the surfaces amalgamated to prevent sticking and held together by slip rings pressed on the slightly tapered outside surfaces.

b) Procedure: Four test specimens shall be prepared in the form of discs 9.5 mm thick and 45 mm diameter. Approximately 115 g of the material sample shall be heated over a low flame until it becomes fluid, then poured into a mold or molds. The sample shall be melted at the lowest possible temperature. The sample shall be stirred thoroughly until it is homogeneous and free from air bubbles when poured into the molds. The material may be poured in a 45 mm diameter cylinder of such length that the four test specimens may be cut from it, using a wire cutter. Two of the samples used in the shock test may be used later to prepare samples for the flow test.

The four specimens shall be cooled in a brine of ice and salt at a temperature of -1°C for a period of at least one hour. The samples shall then be removed one at a time and quickly placed on the anvil of the test apparatus and centered under the plunger. The hammer shall then be tripped from a height of 130 mm. Not more than eight seconds shall elapse from the time each specimen is removed from the brine until the hammer strikes the plunger.

For material to be acceptable, not more than one of the four test specimens shall show a crack. If it is difficult to determine visually whether a specimen has cracked, it may be

removed from the test apparatus and bent slightly.

A2.4 Flow Test

The ability of the coating on the pipe to remain in place and not flow or sag at summer temperatures is indicated by successfully meeting the following flow tests:

a) Apparatus: The apparatus for the flow test shall consist of a corrugated brass plate 0.8 mm thick, 200 mm long and 100 mm wide, with corrugations running the long way of the plate and a metal support to hold the plate on a 45 angle with the horizontal. The corrugations shall have a crest-to crest dimension of 16 mm and depth of 4.8 mm. A line shall be scribed 150 mm from the lower edge. Split molds shall be similar to those specified above for the shock test.

b) Procedure: Two test specimens in the form of cylinders, each 9.5 mm in diameter and 19.0 mm in length, shall be obtained by pouring the excess molten from the shock test into the amalgamated brass molds. Each specimen shall be placed in a corrugation of the corrugated slide (slide to be on the 45° slope) so that the lower end of each specimen will rest exactly along a line scribed 150 mm from the bottom edge of the slide. The test apparatus, with the specimens in place, shall then be placed in an oven maintained at 64°C (± 1°). After 4 hours it shall be removed and allowed to cool to room temperature. The distance from the bottom of the corrugated plate to the lower edge of each test specimen subtracted from 150 mm, determines the amount of sag or flow. To be acceptable, the flow must not exceed 7.0 mm for either of the two specimens.

A2.5 Imperviousness Test

The imperviousness of the coating shall be determined by placing 25 to 50 millilitres each of a 25% solution of sulphuric acid, a 25% solution of sodium hydroxide, and a saturated salt solution, in three adjacent corrugations of the coated pipe or in three corrugations of a section approximately 100 mm wide cut from the pipe. The three solutions shall remain in corrugations for a period of 48 hours during which time no loosening or separation of the coating from the galvanizing shall have taken place.

SECTION 320
CORRUGATED STEEL PIPE

APPENDIX III

Test Method for Lateral Seam Strength of Helical CSP Lock Seam

(Referred to in DBSS 320.09.04.03)

A3.1 Apparatus

a) Testing Machine - Any standard testing machine capable of maintaining the specified rate of strain and of measuring tensile load with an error not to exceed ± 50 N.

b) Grips - The grips for holding the test coupon shall be of the self-aligning type.

A3.2 Preparation of Sample

a) Torch or saw cut a piece approximately 100 mm x 200 mm from the end of the pipe (See Figure 1).

b) Saw cut a 25 mm by 150 mm coupon with parallel sides perpendicular to the lock seam (See Figure 1).

c) Flatten ends of the coupon for gripping in the testing machine. Do this in a vise one end at a time, insert strip in vise to 12 mm from the lock seam. As vise flattens corrugation, strip will rotate. Realign lock seam with flattened ends by use of crescent wrench or vise grips tightened over the lock seam to prevent distortion of the seam itself while bending the strip back into line.

A3.3 Procedure

The gauge length of the test coupon shall be measured in the direction of the seam with calipers to an accuracy of ± 0.2 mm.

The coupon shall be carefully placed in the grips of the testing machine in such manner as to ensure that the axis of the test coupon coincides with the direction of the applied pull.

The load shall be applied at a uniform rate of motion of the movable cross head of 2.5 mm per minute. The maximum load in Newtons shall be recorded.

A3.4 Calculation

The lateral seam strength shall be calculated by the following formula:

$$S = P/L$$

where: S = Lateral Seam Strength in N/mm
P = Maximum Load N
L = Gauge length of test coupon in the direction of the seam in mm

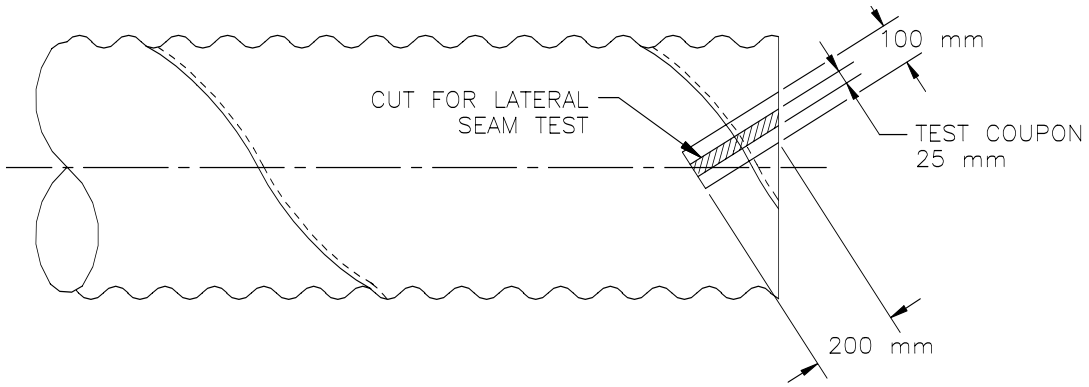


FIGURE 1 - LOCATION OF LOCKSEAM SAMPLE

SECTION 320
CORRUGATED STEEL PIPE

APPENDIX IV

Method of Obtaining Lock Seam Sample from Pipe With Recorrugated Ends

(Referred to in DBSS 320.03.04.04)

A4.1 Tools

The following tools are required:

- Heavy duty 12.5 mm variable speed drill complete with a 38 mm and a 64 mm hole saw.
- Hammer
- Centre punch
- Crescent wrench (200 mm)
- Screwdriver.

A4.2 Procedure

Drill a 38 mm diameter sample with the hole saw. Centre the drill on the centre of the lock seam about 600 mm from the end of the pipe clear of the recorrugated portion.

The sample of the lock seam shall be ground flat across the seam and shall be examined for all the visual inspection requirements contained in DBSS 320.03.04.02.

A4.3 Repair

Repair the pipe by bolting 64 mm zinc coated washers on either side of pipe. See Figure 1. Washers may be obtained by use of the 64 mm hole saw on the tangent of a piece of corrugated sheet.

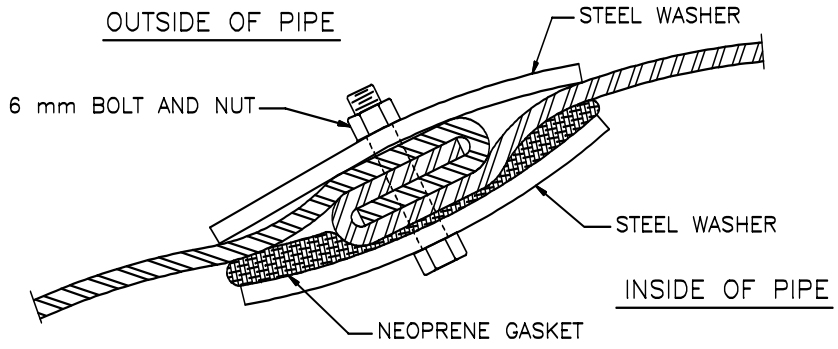


FIGURE 1

SECTION 321

TRAFFIC PAINT

321.01 Scope - This Section applies to paint suitable for spray application at a temperature of 50ºC ± 5ºC to asphalt or concrete pavements for marking of traffic lines. It will be used with overlay glass reflectorized beads, and will be supplied in white or yellow without any "pre-mix" beads.

321.02 Not Used

321.03 General Requirements - The composition of the paint will be at the discretion of the manufacturer but shall be a product pre-qualified by the specified procedure. (See DBSS 321.11 for tendering prerequisites).

321.04 Detail Requirements - Paint to this specification shall comply with the following:

a) Consistency: 85 ± 5 Kreb units. Test method ASTM D 562-01.

b) Drying Time: Maximum 30 minutes at 21ºC at 50-60% relative humidity. Test method ASTM D 711.

c) Bleeding: The paint shall have a degree of resistance to bleeding of seven or greater. Test methods ASTM D 969 and D 868.

d) Road Service Life: The paint shall obtain satisfactory ratings for general appearance, durability and night visibility during seven to eight months of service (June to December approximately) under heavy traffic on bituminous pavement in the vicinity of Victoria or Vancouver, British Columbia. Satisfactory ratings are defined as being equal to or better than those obtained by paints currently on the Ministry's "Recognized Products List"

Test Method: British Columbia Ministry of Transportation Standard Method of conducting and rating Road Service Tests on Traffic Paint (See DBSS 321.11).

e) Handling Qualities:

i) The degree of settling after five months storage shall be such that the paint can be readily re-mixed to a uniform consistency and there shall be no skinning. The paint shall not have marked thixotropic properties.

ii) The paint shall be capable of being sufficiently atomized to produce a uniformly applied line 100 mm in width with absence of side splatter, overspray, or cobwebbing within the limits imposed by the application equipment used on the line striping machines.

iii) The paint shall be capable of being successfully applied to at least 2 km of dash line without the necessity of making adjustments to any machine settings and without showing any evidence of distortion of the spray pattern or appreciable build up of paint in the spray gun tips**.**

Test Method: British Columbia Ministry of Transportation Standard Method of conducting Handling Quality Tests on Traffic Paint (See DBSS 321.11).

f) Colour:

i) White paint shall comply with US Federal Specification 595b White 17886.

ii) Yellow paint shall match British Columbia Ministry of Transportation Standard Yellow Traffic Paint colour chip.

g) Quality Control: Paint supplied under contract shall be identical with the sample submitted for qualification (See DBSS 321.03). Paint shall be considered identical if it has been manufactured within the following tolerances:

i) Mass per litre: ± 25 g of the value on the Traffic Paint Data Form.

ii) Drying time: Maximum 30 minutes and within ± 5 minutes of the value shown on the Traffic Paint Data Form.

iii) Composition of Paint: Within the tolerances indicated by the manufacturer on the Traffic Paint Data Form.

321.05 Packaging - Paint required to be supplied in sealed, vapour-proof containers.

321.06 Not Used

321.07 Release for Shipment

a) The manufacturer shall run quality control tests for each batch of material and shall report the results on the prescribed form.

b) The manufacturer shall notify the Contractor of times at which each batch is to be packaged. Samples shall be taken by the Contractor's Representative at that time.

c) The batch will be released for shipment by the Contractor after sufficient tests have been run to identify the batch with the sample originally qualified and after receipt of the manufacturer's quality control test results. Such release shall not be a bar to subsequent rejection of the material if it is found to have unsatisfactory handling qualities in the field (See DBSS 321.04 (e)) or if its properties of consistency and drying time fail the requirements within one month of the date of manufacture.

321.08 Labelling - Each container shall be clearly marked with Batch Number, Date of Manufacture, Name of Manufacturer and Manufacturer's Code Number on both drums and lids. These markings shall be affixed so that they will not be affected by weather or handling.

321.09 Rejection - Paint rejected for non-compliance with these specifications shall be replaced with paint that does so comply.

321.11 Qualification Procedures and Ministry Test Methods - Details regarding procedures for qualification and Ministry Test Methods may be obtained from the Senior Materials and Pavement Engineer, (phone (250) 387-4360).

Note: Samples for qualification must be received not later than April 1st of each year to obtain qualification for the following year's supply.

SECTION 407

FOUNDATION EXCAVATION

407.01 Scope - This Section covers foundation excavation (i.e., excavation for abutments, piers, culverts, walls, manholes, paving, riprap, etc., not requiring cofferdams), as specified herein, and shall apply generally to all structures other than those in deep water.

407.02 Extent of Work - Foundations shall be excavated to the lines, levels and grades established by the Designer. All organic and soft materials shall be entirely removed to a firm formation and piled well back from the work. The use of explosives shall be limited strictly to the necessities of the job. As nearly as practicable, the foundation provided shall be of uniform bearing capacity.

407.03 Not Used

407.04 Not Used

407.05 Not Used

SECTION 412

REINFORCING STEEL

DESCRIPTION

412.01 Scope - This Section describes the supply, fabrication and installation of reinforcing steel for concrete structures.

The requirements of CAN/CSA-S6-00, "Canadian Highway Bridge Design Code" shall apply unless specified otherwise herein or in the Contract.

For items not covered by CAN/CSA-S6-00, "Canadian Highway Bridge Design Code" or the Contract, CAN/CSA A23.1 shall apply.

References to standards shall be to the current editions at time of Award.

MATERIALS

412.11 General - All reinforcing bars shall be deformed except for spirals and ties where plain bars may be used.

The type and grade required shall be as specified on the Design.

Wire ties for reinforcing steel, conforming to CSA G30.3, shall be minimum 1.6 mm diameter, cold drawn, annealed steel wire.

Two copies of mill certificates giving chemical and physical properties of the reinforcing steel shall be furnished to the Quality Manager and, when requested, to the Ministry Representative.

The identification of reinforcing bars shall be maintained throughout the fabrication, coating (if required) and shipping processes to the installation on the job.

TABLE 412-A REQUIREMENTS FOR UNCOATED REINFORCING STEEL

G30.3	Cold-Drawn Steel Wire for Concrete Reinforcement
G30.5	Welded Steel Wire Fabric for Concrete Reinforcement
G30.14	Deformed Steel Wire for Concrete Reinforcement
G30.15	Welded Deformed Steel Wire Fabric for Concrete Reinforcement
G30.18	Billet Steel Bars for Concrete Reinforcement

The Contractor shall as part of its quality control program have random samples (selected by the Quality Manager or Ministry Representative) of reinforcing steel, both coated and uncoated, tested for bending and tension. Two samples shall be taken of each size and grade of bar used on the project. Testing shall be in accordance with CAN/CSA-G30.18-M92 unless otherwise specified. Testing shall be performed by an approved testing agency.

412.11.01 Uncoated Reinforcing Steel - Concrete reinforcement shall conform to the requirements of the CSA Standards shown in Table 412-A, as applicable.

412.11.02 Epoxy-Coated Reinforcing Steel - Epoxy-coated reinforcing steel bars shall conform to the requirements of ASTM A-775M and D-3963M. Application plants shall be certified under the Concrete Reinforcing Steel Institute (CRSI) Voluntary Certification Program for Fusion-Bonded Epoxy-Coating Applicator Plants, and proof of certification shall be submitted to the Quality Manager and, when requested, to the Ministry Representative. Pre-treatment of the near white blast cleaned surfaces with chromate or equivalent is mandatory prior to the application of a minimum of 180 μm of cured thickness of protective epoxy-coating by an electrostatic spray method. All epoxy-coated bars shall be clearly labelled with the name of the manufacturer, the primer and the coating system used and the date of production. The method of labelling shall be determined by the applicator plants and test certificates of performance of coated bars shall be submitted to the Quality Manager and, when requested, to the Ministry Representative.

Further to paragraph 9.2 of ASTM D-3963, the identification marks on all coated reinforcing bars shall be maintained from the point of delivery until they are incorporated into the structure.

The Quality Manager or Ministry Representative may require samples of epoxy resin and epoxy patching material for acceptance testing as called for in ASTM A-775M. The Quality Manager or Ministry Representative may require tests of coated bars as called for in ASTM A-775M.

412.11.03 Galvanized Reinforcing Steel – Galvanized reinforcing bars shall be used only where shown on the Design. Substitution of galvanized reinforcing bars in place of epoxy-coated reinforcing bars will not be permitted. Substitution of epoxy-coated reinforcing bars in place of galvanized

reinforcing bars will not be permitted. Hot dipped galvanized reinforcing bars shall conform to the requirements of ASTM A-767M – *Class 1 coating.*

Only Grade W bars shall be galvanized

Galvanized reinforcing bars shall not be placed in contact with uncoated reinforcing bars.

412.11.04 Fibre Reinforced Polymer (FRP) - FRP components and FRP reinforcing bar shall conform to CSA Specification S806. FRP reinforcing bars shall be used only where shown on the Design.

412.11.05 Order Lists and Bending Diagrams - All order lists and bending diagrams shall be submitted to the Designer and if requested, the Ministry Representative for review before material is ordered.

The review of order lists and bending diagrams by the Ministry Representative shall not relieve the Contractor or Suppliers, as applicable, of responsibility for the correctness thereof.

TABLE 412-B MINIMUM INSIDE DIAMETERS OF BENDS*

BAR DESIG-NATION	UNCOATED BARS (mm)			EPOXY COATED BARS (mm)
	300R	400R or 500R	400W or 500W	
10M	60	70	60	80
15M	90	100	90	120
20M	-	120	100	160
25M	-	150	150	200
30M	-	250	200	240
35M	-	300	250	350
45M	-	450	400	450
55M	-	600	550	550

CONSTRUCTION

412. 31 Bending - Reinforcement bars shall be cut and bent to the shapes shown on the Design. Bending shall be sufficiently accurate that the placing tolerances (DBSS 412.33.01) can be met. All bars shall be bent cold, unless otherwise permitted by the Designer and the Ministry Representative. Bars partially embedded in concrete shall not be field bent except as shown on the Design or as specifically permitted by the Designer and the Ministry Representative.

412.31.01 Hooks and Bend Dimensions - Where hooks are shown on the Design, they shall have the following dimensions, unless shown otherwise:

- 180° bend plus extension of at least 4 bar diameters, but not less than 60 mm.
- 90° bend plus extension of at least 12 bar diameters.
- for stirrups and ties only, either a 90° or a 135° bend plus extension of at least 6 bar diameters.at the free end of the bar.

- Minimum inside diameters of bends shall be as shown in Table 412-B.

Exceptions:

- Minimum inside diameters of bends and 90º and 135º hooks for stirrups and ties shall be 4 bar diameters for uncoated bars and 8 bar diameters for epoxy-coated bars.
- Minimum inside diameters of bends in welded wire fabric, plain or deformed, for stirrups and ties shall not be less than 4 wire diameters for deformed wire larger than 7 mm and two wire diameters for all other wires, except that bends with an inside diameter of less than 8 wire diameters shall be not less than 4 wire diameters from the nearest welded intersection.

- Note: Reinforcing bars shall be pre-bent before galvanizing unless acceptable to the Quality Manager and Ministry Representative. Minimum bend diameters if pre-bent and galvanized will be as for uncoated bars.

Bars partially embedded in concrete shall not be field bent except as permitted by the Designer and the Ministry Representative.

412.32 Handling and Storage - Reinforcing bars shall be stored on platforms, skids or other suitable supports clear of the ground and shall be protected as practicable from mechanical injury and surface deterioration caused by exposure to conditions producing rust. When placed in the work, reinforcement shall be free from dirt, loose rust or scale, mortar, paint, grease, oil, or other materials that would reduce bond.

412.32.01 Epoxy-Coated Reinforcing Steel - Epoxy-coated reinforcing bars shall be handled and stored so as to minimize damage to the coating. Equipment for handling the bars shall have protected contact areas. Nylon slings or padded wire rope slings shall be used. Suitable bundling bands shall be used to prevent damage. Bundles of coated bars shall be lifted at multiple pick-up points to prevent bar-to-bar abrasion from sags in the bundles of bars. The bars or bundles

shall not be dropped or dragged.

Coated steel reinforcing bars shall be stored off the ground on protective cribbing, and timbers placed between bundles when stacking is necessary. Supports shall be placed sufficiently close to prevent sags in the bundles.

Long term storage of epoxy-coated bars shall be minimized
If circumstances require storing coated steel reinforcement outdoors for more than two months, protective storage measures shall be implemented to protect the material from sunlight, salt spray and weather exposure. Coated steel reinforcing bars stored in corrosive environments will require protection sooner. Coated steel reinforcing bars or bundles shall be covered with opaque polyethylene sheeting or other suitable opaque protective material.

Coated bars shall not be cut or bent on the job without the permission of the Quality Manager and Ministry Representative. In any case, the flame-cutting of coated bars will not be permitted.

When placing coated steel reinforcing bars, all wire bar supports, spacers, and tying wire shall be coated with non-conductive material, for example, an epoxy-coated or plastic coated material compatible with concrete.

When immersion-type vibrators are used to consolidate concrete around epoxy-coated steel reinforcing bars, the vibrators shall be equipped with rubber or non-metallic vibrator heads

Any bar with coating damage exceeding 2 % of the surface area of the coated steel reinforcing bar in any 300 mm length shall be rejected. When the extent of the damage does not exceed 2 % of the surface area in any 300 mm length, all damaged coating shall be repaired with patching material in accordance with ASTM A 775M, Section 12. Patching material shall be applied in strict accordance with the manufacturer's written instructions. Prior to application of the patching material, rust shall be removed from the damaged areas. The patching material shall be allowed to cure before placing concrete over the coated steel reinforcing bars.

The "Guidelines for Jobsite Practices", contained as an Appendix of ASTM A 775M shall be mandatory.

Where on-site inspection or testing by the Quality Manager or Ministry Representative does not confirm compliance of the coating thickness or integrity, for any shipment of bars, such bars may be rejected, and in any case shall not be placed in the structure until the Quality Manager and Ministry Representative are satisfied as to their acceptability.

The Contractor shall make available to the Quality Manager and Ministry Representative, for inspection and acceptance, all repairs performed to protective coatings on any reinforcing steel. Repairs of epoxy coatings shall not be performed when the coated bar or ambient air temperature is 5°C or less, or when moisture is present on the bar. Patching material shall not be applied if precipitation is expected within four hours of the procedure. If precipitation does occur within four hours, the Quality Manager shall determine as to whether the material as applied is acceptable or must be replaced. If the placed coated bars have been exposed to salt spray or road salts, the bars shall be rinsed with fresh water to remove any chloride contamination prior to placing concrete.

412.33 Placing and Fastening - Before any concrete is placed, the placing and securing of reinforcing steel including dowels, within the area of concrete placement shall be complete. Tying in place of all dowels projecting from the area of concrete placement shall be acceptable to the Quality Manager before any concrete is placed. The reinforcing steel shall be free from dirt, detrimental rust, loose scale, paint, oil or other foreign material.

Reinforcement shall be placed in the positions shown on the Design, within the tolerances specified below, adequately supported and secured against displacement. All splices of adjacent bars shall be securely tied together. Tying 100% of the bar intersections is required where the bar spacing centres are 300 mm or greater. Tying of 50% of the bar intersections is required when the bar spacing centres are less than 300 mm.

The locations of the top reinforcing steel in bridge decks shall be checked by running a full deck-width template along the longitudinal screeds. The lower edge of the template shall be set at the nominal cover dimension (reference the Design) below the level of the deck surface. No steel shall touch the template nor be more than 6 mm distant from it.

The location of reinforcing steel near deck joint anchors shall be adjusted so that there will be no interference with the deck joint anchors.

Tack welding of reinforcement for cage assembly or securing of reinforcement will be permitted only with the written acceptance of the Designer. In no cases shall stirrups be welded to tensile reinforcement.

412.33.01 Tolerances - Tolerances for placing reinforcement shall, unless otherwise specified, be as